Mohamed Shehata

AI Engineer

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Portfolio

PROFILE

AI Engineer with 2+ years of experience shipping production LLM systems: agentic workflows, RAG, and multimodal pipelines spanning speech, vision, and documents. Specialized in on-premise model serving. Graduate of the 9-month ITI AI and Machine Learning Engineer diploma, with a B.Sc. in Data Science (GPA 3.68/4.0, Excellent).

EDUCATION

9-Month Professional Diploma : AI And Machine Learning Engineer Track 10/2023 – 07/2024
Information Technology Institute (ITI), MCIT - Alexandria

Bachelor's Degree In Data Science 09/2019 – 06/2023
Faculty of computers and data sciences, Alexandria University.
GPA: 3.68/4.0 — *Graduated with Excellent Grade*

PROFESSIONAL EXPERIENCE

AI Engineer 01/2026 – Present
wakeb_Data Cairo, Egypt

- Built **AI agents and chatbots** that automate end-to-end application workflows.
- Built a **Report & Presentation Generation Agent** that, from raw data or deep web research across 100+ sources, generates business-ready reports and 200+ page slide decks in 10–60 seconds.
- Developed an **Intelligent Meeting Analysis** Platform that processes up to 3-hour meetings in ~5 minutes: **transcription, speaker diarization,** entity extraction, structured summaries, and actionable insights.
- Developed a Multimodal Generation System that **generates and edits images, creates videos**, and produces audio, with all models self-hosted on vLLM-Omni.
- Built an Intelligent Document Processing Platform **combining OCR, document understanding,** summarization, and image captioning to convert complex documents into structured, actionable data.
- Deployed and optimized local **LLMs in on-premise** production environments, enabling scalable, low-latency inference.

AI & Machine Learning Engineer 11/2024 – 01/2026
intalio Cairo, Egypt

- Built RAG pipelines, **SQL agents,** and **interactive chart-generation** agents with ~3-second response times.
- **Fine-tuned large language models** for domain-specific tasks, enabling CPU-based inference with ~5-second response times on summarization.
- Designed end-to-end multimodal pipelines integrating **OCR**, image–text retrieval, document intelligence, **text-to-speech**, and automatic **speech recognition** into unified systems.
- Developed and fine-tuned deep learning models for an end-to-end **Egyptian and Lebanese national ID processing pipeline**, integrating ID detection, OCR, and structured field extraction, achieving **~1-second GPU inference latency per ID**.
- Built interactive **conversational AI avatars** combining LLMs with STT/TTS, achieving ~5-second response latency on GPU.

AI Engineer Intern 06/2024 – 09/2024
TwentyToo.ai Remote

- Built three fashion recommendation systems: visual similarity, personalized recommendations, and frequently-bought-together products.
- Implemented Fashion CLIP, Two-Tower models, and FP-Growth to improve product discovery, personalization, and cross-selling

TECHNICAL SKILLS

LLMs & Agents: LangGraph, LangChain, CrewAI, DSPy, Instructor, MCP

RAG: Vector Databases (Qdrant, FAISS), Embeddings (BGE-M3, Qwen, Cohere), Evaluation (RAGAS), Hybrid Search, Reranking, Chunking Strategies

LLMOps, Evaluation & Reliability: LangSmith, Langfuse, DeepEval, LLM-as-a-Judge, Guardrails, Human-in-the-Loop

ML & Data: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, SQL, Plotly, Streamlit

Multimodal AI: Image & Video (OCR, Captioning, Generation, Editing), Speech (ASR, Diarization, TTS).

Model Serving & Optimization: vLLM, vLLM-Omni, llama.cpp, Ollama, Hugging Face Transformers, LoRA Fine-Tuning (Unsloth, LLaMA-Factory), Quantization.

Backend & MLOps: Python, FastAPI, Docker, Redis, Celery, AWS, Linux, Git, CI/CD, Bash, Prometheus, Grafana

Core Concepts: Agentic AI, Multi-Agent Orchestration, RAG, LLM Applications, Recommendation Systems, Tool/Function Calling, Computer Vision, NLP

PROJECTS

Blind Assistance System (Faculty Graduation Project)
- Developed a solution using smart glasses and a mobile app for blind individuals, incorporating object detection, text reading, scene description, cash/food recognition, and face recognition.
- **Research paper :**Smart Equipment for Assisting Blind People Published in (IJCSIS),Vol. 21, No.7 Jul
- Tools**: TensorFlow,** PyTorch, OpenCV.

Necromancer: Colorization and Captioning for Videos and Images (ITI)
- Used DeOldify for artistic colorization of black-and-white media into vibrant images and videos. Implemented Salesforce's BLIP model for contextual captions and integrated audio using gTTS for enhanced multimedia experience
- Tools**: Transformers, DeOldify, Opencv, torch, gTTS, Streamlit**

TalkAgent: Voice-Enabled Customer Service Agent
- Built a LangGraph customer service agent powered by Gemini, with RAG over PDFs (FAISS) and escalation to a human agent.
- Added CRM tools for appointments and complaints, and ElevenLabs voice chat.
- Tools: LangGraph, Gemini, FAISS, SQLite, ElevenLabs

AI-Powered Exam Monitoring and Proctoring Platform
- Developed an advanced AI-powered exam monitoring platform that integrates facial recognition and verification, eye movement tracking, lip movement analysis, voice detection, object detection, and activity monitoring.
- Tools : PyTorch, MediaPipe, OpenCV, Pyaudio and Flask.

IMDb Analysis and recommendation system For Top Movies and Series (ITI)
- Developed end-to-end IMDb analysis pipeline, web scraping, data preprocessing , exploratory data analysis, dashboard development, recommendation system.
- **Featured on Plotly Official Page**
- Tools: Selenium, Plotly Dash, Pandas, Scikit-learn

COURSES

- **Data Scientist Professional with Python (116 hours) - DataCamp**
- **Data Analyst in Power BI (48 hours) - DataCamp**
- **Data Analyst in SQL (39 hours) - DataCamp**
- **Data Analyst in Tableau (42 hours) – DataCamp**
- **Advanced Data Analysis - Udacity**
- **AWS Cloud Foundations - AWS Academy**

LANGUAGES

- Arabic: Native
- English: Very Good